Responses to the comments of the reviewer 1:

General Response: We would like to thank you for your efforts to review our manuscript. We were greatly encouraged by your helpful comments to improve our manuscript. Our answers are inserted right after your comments. The correspondingly added or modified texts are marked with pink color in the revised manuscript.

This work reports a $g\text{-}C_3N_4$-assisted solid-state strategy for fabricating nitrogen-doped carbon fabrics with tunable nitrogen configurations. The topic aligns well with the journal's scope. The experimental design is generally sound, and the combination of experimental characterization with DFT calculations strengthens the mechanistic insight. However, several critical issues regarding nitrogen speciation accuracy, performance benchmarking, and practical applicability need to be addressed.

The manuscript presents a valuable and timely study, but requires additional experiments and clarifications to fully support its conclusions before publication.

Comments:

C1. The overlap between residual $g\text{-}C_3N_4$ C=N-C (~398.1 eV) and pyridinic-N (~398.3-398.8 eV) in XPS makes it difficult to accurately quantify lattice-incorporated pyridinic nitrogen, especially in NCF-600 and NCF-700. Please explain.

Response: We thank the reviewer for this insightful comment. We agree that the close proximity of the N 1s binding energies for $g\text{-}C_3N_4$ C=N–C moieties (~398.1 eV) and lattice-incorporated pyridinic-N (~398.3–398.8 eV) poses an inherent challenge for unambiguous deconvolution by XPS alone.

To clarify this issue and improve the accuracy of our interpretation, we have explicitly addressed this limitation in the revised manuscript by correlating the XPS analysis with the TGA results. For the low-temperature samples (NCF-600 and NCF-700), TGA indicates incomplete decomposition of the $g\text{-}C_3N_4$ precursor. Therefore, the apparent high pyridinic-N fractions observed in these samples may partially originate from residual $g\text{-}C_3N_4$-derived C=N species rather than exclusively from electrochemically active lattice-incorporated pyridinic-N sites.

In particular, NCF-600 represents the most extreme case. Although a significant signal is observed in the pyridinic-N region, the coexistence of multiple characteristic $g\text{-}C_3N_4$ nitrogen species suggests that a substantial fraction of the nitrogen remains associated with the polymeric $g\text{-}C_3N_4$ framework rather than being incorporated into a conductive carbon lattice. NCF-700 represents a transitional state in which $g\text{-}C_3N_4$ decomposition is nearly completed, but the nitrogen species have not yet fully reconstructed into electrochemically effective lattice configurations.

For samples carbonized at higher temperatures, the contribution from residual g-C_3N_4-derived species is expected to become progressively smaller due to the continued thermal decomposition and reconstruction of nitrogen species within the carbon framework. Consequently, the signal at 398.3–398.8 eV in these samples can be more reasonably attributed to lattice-incorporated pyridinic-N.

On pages 9 – 10, Section 3.1 of the revised manuscript, we have added and revised the discussion as follows:

"….The presence of these multiple g-C_3N_4-related nitrogen configurations indicates incomplete thermal decomposition at 600 °C, consistent with the TGA results [25]. Under these conditions, a substantial fraction of the nitrogen species remains associated with the polymeric g-C_3N_4 framework rather than being incorporated into a conductive carbon lattice. Consequently, part of the ~398 eV signal in NCF-600 is likely associated with residual g-C_3N_4-derived C=N moieties. Therefore, the high pyridinic-N fraction observed in NCF-600 is attributed, at least in part, to residual g-C_3N_4-derived species rather than exclusively to lattice-incorporated pyridinic-N. Although NCF-700 exhibits a relatively high fraction of pyridinic-N according to XPS analysis….

"…making it difficult to unambiguously distinguish lattice-incorporated pyridinic-N from residual g-C_3N_4-derived species by XPS alone. Therefore, the elevated pyridinic-N fraction observed in low-temperature samples does not directly reflect the density of structurally integrated pyridinic sites within the carbon framework. Instead, only pyridinic-N species effectively incorporated into the carbon lattice are expected to contribute significantly to the functional properties of the material. For samples pyrolyzed…"

These revisions-r clarify the uncertainty associated with XPS-based quantification of pyridinic-N in NCF-600 and NCF-700 and provide a more rigorous interpretation of the nitrogen species present in low-temperature samples.

C2. The manuscript lacks direct performance comparison with recently reported nitrogen-doped carbon fabrics or graphene-based materials. The influence of electrolyte pH on pyridinic-N activity is not explored.

Response: We thank the reviewer for this valuable suggestion. First, to provide a clearer assessment of the electrochemical performance of the developed material, we have added a new comparison table summarizing the performance of recently reported nitrogen-doped carbon fabrics and related nitrogen-doped carbon-based electrodes for supercapacitor applications. The comparison demonstrates that the NCF-900 electrode exhibits competitive capacitance and cycling stability compared with previously reported systems.

A new comparison table (Table S2) and the corresponding discussion have been added **on pages S-12 – S-13** of the revised support information.

"Table S2. Comparison of the electrochemical performance of g-C_3N_4-derived NCF-900 with recently reported nitrogen-doped carbon-based electrodes for supercapacitor applications.

Material	Nitrogen Source / Strategy	Electrolyte	Capacitance	Cycling	Ref.
N-doped carbon fabrics	Melamine, cotton fabric / Pyrolysis	6 M KOH	180 F g^{-1} @ 0.5 A g^{-1}	95% (5,000 cycles)	[S1]
N-doped porous carbon	Juncus, $ZnCl_2$ / Pyrolysis	6 M KOH	290.5 F g^{-1} @ 0.5 A g^{-1}	94.5% (10,000 cycles)	[S2]
N-doped carbon	PAN, melamine / Pyrolysis	6 M KOH	135.3 F/g @ 0.5 A g^{-1}	96.6% (5,000 cycles)	[S3]
N-doped carbon	Citric acid, urea / Wet ball milling	1 M Na_2SO_4	79.25 F g^{-1} @ 1 A g^{-1}	96.1% (2,500 cycles)	[S4]
N-doped porous carbon	$ZnCl_2$, NH_4Cl, Ginger straw / Pyrolysis	EMIM-BF_4	122 F g^{-1} @ 0.5 A g^{-1}	87% (10,000 cycles)	[S5]
N-doped porous carbon	Walnut shells, melamine / KOH activation	6 M KOH	329.2 F g^{-1} @ 1 A g^{-1}	90.12 % (5000 cycles)	[S6]
N-doped carbon fabrics	**g-C_3N_4, cotton fabric / Pyrolysis**	**1 M H_2SO_4**	**789.4 mF cm^{-2} @ 1 mA cm^{-2}**	**69.6 % (5000 cycles)**	**This work**

N-doped carbon fabrics	**g-C_3N_4, cotton fabric / Pyrolysis**	**6 M KOH**	**139.6 F g^{-1} @ 0.5 A g^{-1}**	**89% (10,000 cycles)**	**This work**

5. Supporting references

[S1] Chen, L., Ji, T., Mu, L., & Zhu, J. (2017). Cotton fabric derived hierarchically porous carbon and nitrogen doping for sustainable capacitor electrode. Carbon, 111, 839-848.

[S2] He, G., Yan, G., Song, Y., & Wang, L. (2020). Biomass Juncus derived nitrogen-doped porous carbon materials for Supercapacitor and oxygen reduction reaction. Frontiers in Chemistry, 8, 1.

[S3] Fu, X., Li, R., Yan, S., Yuan, W., Zhang, Y., Sun, W., & Wang, X. (2023). Porous carbons prepared from polyacrylonitrile doped with graphitic carbon nitride or melamine for supercapacitor applications. ChemistrySelect, 8, e202301801.

[S4] Devi, M., Shikhar, J., & Sharma, S. (2025). Effect of nitrogen content on performance of supercapacitors composed of nitrogen–carbon materials. Journal of Materials Chemistry A, 13, 42343-42354.

[S5] Zheng, Y., Wang, H., Sun, S., Lu, G., Liu, H., Huang, M., ... & Li, H. (2020). Sustainable nitrogen-doped carbon electrodes for use in high-performance supercapacitors and Li-ion capacitors. Sustainable Energy & Fuels, 4, 1789-1800.

[S6] Niu, C., Ma, X., Zhou, J., Shi, Q., Qu, H., Feng, X., ... & Zhang, R. (2025). Controlled synthesis of pyridinic-enriched nitrogen-doped and high specific surface area porous carbon derived from walnut shell for supercapacitors. Chemical Engineering Science, 122499."

On page 15, Section 3.2 of the revised manuscript, we have added the sentences as follows:

"Overall, these results highlight... A benchmark comparison with representative nitrogen-doped carbon-based electrodes is summarized in Table S2. Although direct comparison is influenced by differences in testing conditions and electrolytes, NCF-900 exhibits competitive electrochemical performance and cycling stability among recently reported nitrogen-doped carbon materials."

Regarding the influence of electrolyte pH, we agree that the electrolyte environment can significantly affect the electrochemical activity of nitrogen-containing functional groups. Although a systematic pH-dependent study was not the primary objective of this work, we evaluated the electrochemical response in both acidic (1 M H_2SO_4, pH ≈ 0) and alkaline (6 M KOH, pH ≈ 14) electrolytes. NCF-900 exhibits more pronounced pseudocapacitive behavior in the acidic electrolyte, which is consistent with the proposed proton-coupled charge-storage mechanism. This observation is further supported by the DFT calculations, which reveal a highly favorable proton adsorption energy (Eads = −2.58 eV) for pyridinic-N sites, indicating that these edge-associated nitrogen configurations can effectively participate in reversible proton adsorption/desorption processes under acidic conditions.

On page 12, Section 3.2 of the revised manuscript, we have revised the sentences as follows: "The capacitive behavior of the NCF-T materials was examined using both a three-electrode configuration in 1 M H_2SO_4 electrolyte and a symmetric two-electrode configuration in 6 M KOH electrolyte. In the symmetric two-electrode configuration using 6 M KOH electrolyte, the cyclic voltammetry (CV) curves maintained nearly rectangular profiles at 5 mV s^{-1}, indicative of efficient ion transport and reversible adsorption processes (Figure 4a). In contrast, the CV curves obtained in 1 M H_2SO_4 exhibited broader redox features (Figure 4b), suggesting an additional pseudocapacitive contribution associated with nitrogen-containing functional groups. Among the series, NCF-900 consistently exhibited the highest capacitance in both systems, suggesting that its favorable nitrogen configuration and porous structure are beneficial for charge storage. Corresponding charge–discharge profiles (Figure S6a) revealed longer discharge times for NCF-900 under identical current densities, whereas NCF-600, NCF-700 and NCF-1000 showed lower capacitance, likely due to incomplete nitrogen incorporation into electrochemically active lattice configurations (NCF-600 and NCF-700) or excessive graphitization (NCF-1000)."

On page 13, Section 3.2 of the revised manuscript, we have revised the sentences as follows: "In the three-electrode configuration using 1 M H_2SO_4 electrolyte, the CV curves exhibited broad redox humps superimposed on the capacitive background (Figure 4c), indicating the contribution of pseudocapacitive charge-storage processes associated with nitrogen-containing functional groups [39], [40]. The charge storage…"

Except for the above revisions, we have revised and added some sentences and figures in the revised manuscript. So, we would be very pleased if you take a close look at the revised manuscript. Once more, thank you very much for your really helpful comment.

C3. Real-device demonstration is missing. Demonstrate a flexible pouch cell or a simple application (e.g., powering an LED).

Response: We thank the reviewer for this valuable suggestion. To demonstrate the practical applicability of the fabricated electrodes, we have performed an additional device-level demonstration using symmetric NCF-900 supercapacitors connected in series to power a commercial red LED.

As shown in the newly added Figure 6, one, two, and three symmetric NCF-900 cells were connected in series to increase the operating voltage. The CV and GCD results of the series-connected devices confirm the expected voltage expansion while maintaining stable capacitive behavior. Furthermore, three cells connected in series were successfully used to power a red LED. The LED remained illuminated for more than 3 minutes after charging, demonstrating the ability of the assembled devices to store and deliver electrical energy for practical applications. These results provide direct evidence of the feasibility of the NCF-900 electrodes for real-world energy-storage applications beyond laboratory-scale electrochemical testing.

On page 7, Section 2.6 of the revised manuscript, we have revised the sentences as follows:

2.6. Fabrication and demonstration of symmetric supercapacitor device

The symmetric supercapacitor device was assembled using two circular NCF-900 electrodes (14 mm diameter) directly punched from the carbon fabric and used without any conductive additives or polymer binders. A separator soaked in 6 M KOH electrolyte was placed between the two electrodes, and the assembly was packaged in a coin-cell configuration.

A new **Figure 6** and a **Movie S1** have been added to the revised manuscript, together with the following discussion **on pages 15 – 16**, Section 3.2 in the revised manuscript:

"To demonstrate the practical applicability of the NCF-900 electrodes, symmetric supercapacitor devices were connected in series and evaluated at the device level. As shown in Figure 6a, 6b, the operating voltage increased proportionally with the number of connected cells while maintaining capacitive behavior. Furthermore, three NCF-900 devices connected in series were able to power a commercial red LED for over 3 min after charging (Figure 6c and Movie S1), demonstrating the feasibility of the fabricated electrodes for practical energy-storage applications."

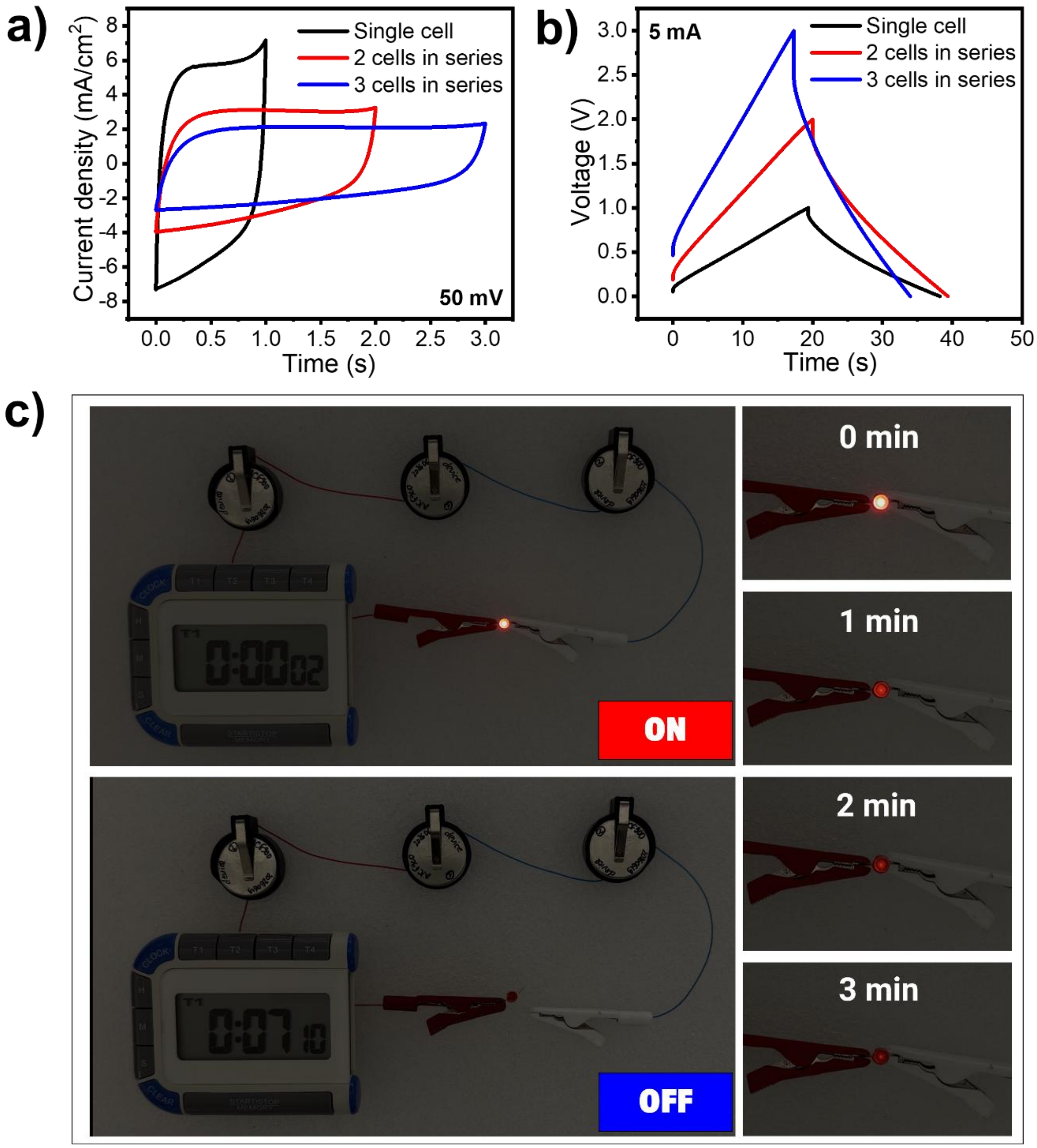

Figure 6. Device-level performance and practical demonstration of NCF-900 symmetric supercapacitors connected in series. **a)** CV curves and **(b)** GCD profiles of one, two, and three cells connected in series; **c)** Photographs demonstrating the illumination of a red LED powered by three series-connected NCF-900 devices and the corresponding discharge process.

Responses to the comments of the reviewer 2:

General Response: We would like to thank you for your efforts to review our manuscript. We were greatly encouraged by your helpful comments to improve our manuscript. Our answers are inserted right after your comments. The correspondingly added or modified texts are marked with pink color in the revised manuscript.

This work provided a solid-state strategy that employs exfoliated g-C_3N_4 nanosheets as both a nitrogen source and porogen to synthesize nitrogen-doped carbon fabrics. The relative proportions of pyridinic, pyrrolic, and graphitic nitrogen species were systematically tuned. An optimal balance of pyridinic-N content, abundant defect sites, and a well-developed mesoporous network can achieve high specific capacitance, excellent rate capability, and capacitance retention after 10,000 cycles. According to the experiment and theory results, the authors concluded that pyridinic nitrogen acted as a major contributor to redox activity by facilitating reversible proton-coupled electron transfer and enhanced ion adsorption. However, there are some issues should be considered.

Comments:

C1. There has always been argument over the relationship between nitrogen configurations doped in carbon structures and the charge storage performance. This study analyzes the relative contents of various nitrogen configurations using XPS results. Other characterization techniques are required to supplement the experiments, as the limited detection depth of XPS means that the analyzed nitrogen configurations only represent the surface phase.

Response: We thank the reviewer for this valuable comment. We agree that XPS is inherently a surface-sensitive technique and therefore does not directly represent the bulk composition of the material. However, the objective of the present study is not to quantify the bulk nitrogen distribution throughout the carbon fibers, but rather to investigate the role of electrochemically accessible nitrogen species generated through a g-C_3N_4-derived surface modification strategy. In our synthesis route, exfoliated g-C_3N_4 nanosheets were intentionally coated onto the cotton fabric prior to pyrolysis. Therefore, nitrogen functionalities are expected to be concentrated near the outer region of the carbon fibers, where electrochemical reactions predominantly occur.

To further address the reviewer's concern, we have added cross-sectional EDS characterization of NCF-900 in the revised manuscript (Figure S1c). The cross-sectional analysis reveals a clear nitrogen signal in the outer region of the carbon fiber, whereas the nitrogen signal becomes significantly weaker toward the interior and is nearly absent in the fiber core. This observation confirms that the nitrogen species are primarily enriched in the surface and near-surface regions rather than being homogeneously distributed throughout the bulk.

Because charge storage in supercapacitors is governed mainly by surface-accessible active sites, the nitrogen configurations identified by XPS remain highly relevant for understanding the electrochemical behavior of the NCF electrodes. Accordingly, we have revised the manuscript to clarify that the conclusions of this work focus on electrochemically accessible nitrogen species rather than the bulk nitrogen composition.

On page 4, section Introduction of the revised manuscript, we have revised the sentences as follows:

"As a result, nitrogen can be introduced in a more controllable manner while preserving the structural integrity of the carbon framework. Nitrogen introduced via this route typically enables better control over the formation of different nitrogen configurations, including pyridinic and graphitic configurations [15]. […] The sample prepared at 900 °C (NCF-900) exhibited the most favorable combination of a high density of electrochemically accessible pyridinic-N sites, defect-rich structure, and mesoporosity, resulting in superior capacitive performance and durability."

On page 8, Section 3.2 of the revised manuscript, we have revised and added the sentences as follows:

"… using exfoliated g-C_3N_4 nanosheets as both the nitrogen source and sacrificial template. The physical appearance of the samples is shown in Figure S1a, while EDS mapping confirms the successful incorporation of nitrogen species within the carbon fabric (Figure S1b). To further examine the spatial distribution of nitrogen species within the carbon fibers, cross-sectional EDS analysis was performed on NCF-900 (Figure S1c). A distinct nitrogen signal was detected near the outer region of the fiber, while the nitrogen intensity decreased markedly toward the interior and became negligible in the fiber core. This result indicates that the nitrogen functionalities introduced by the g-C_3N_4-assisted treatment are predominantly concentrated in the surface and near-surface regions of the carbon fibers. Thermogravimetric analysis…"

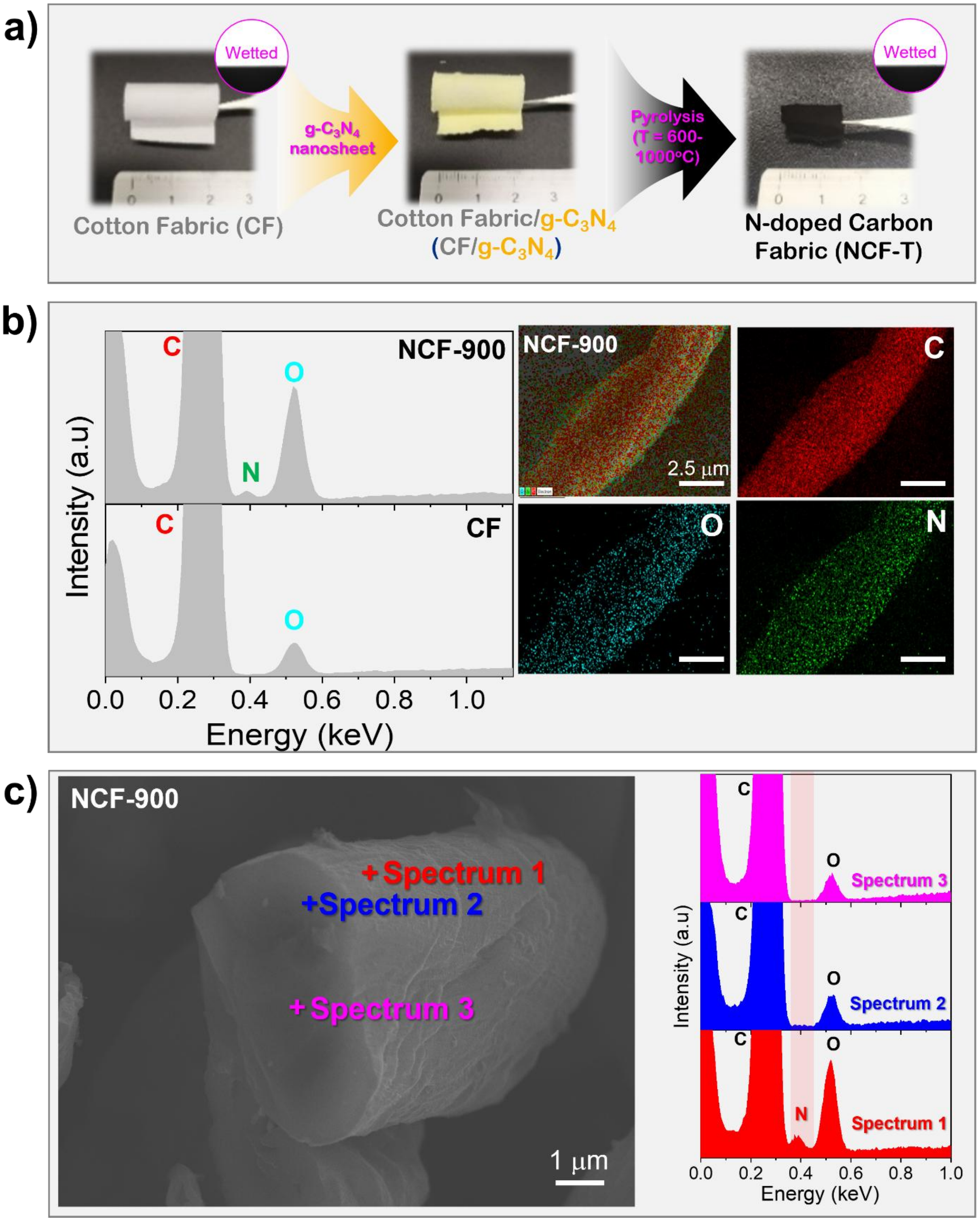

Figure S1. a) Photographs illustrating the fabrication process from pristine cotton fabric (CF) to g-C_3N_4-coated cotton fabric (CF/g-C_3N_4) and nitrogen-doped carbon fabric (NCF-T); **b)** EDS spectra and elemental mapping images of NCF-900; **c)** Cross-sectional SEM image and corresponding EDS point spectra acquired at different positions across the NCF-900 fiber.

C2. Furthermore, the research did not provide data on the overall nitrogen content in the carbon. If the total nitrogen content is low, the effect of nitrogen configurations on performance may not be the primary factor, as the BET data indicate that the specific surface areas of the materials differ as well.

Response: We thank the reviewer for this insightful comment. We agree that both the overall nitrogen content and the specific surface area contribute to the electrochemical performance of nitrogen-doped carbon materials. Therefore, the charge-storage behavior cannot be attributed to a single parameter alone.

Regarding the overall nitrogen content and the actual amount of each nitrogen configuration, we agree that distinguishing between the relative fractions of nitrogen configurations and their actual contents is important for interpreting the electrochemical behavior. Therefore, Table 2 and Table 3 have been explicitly discussed in the revised manuscript. While Table 2 summarizes the relative fractions of pyridinic, pyrrolic, graphitic, and oxidized nitrogen species obtained from N 1s peak deconvolution, Table 3 presents the corresponding total nitrogen contents and calculated atomic contents of each nitrogen configuration. These data reveal that the electrochemical performance does not directly follow either the total nitrogen content or the calculated pyridinic-N content alone. The results show that the total nitrogen content decreases progressively with increasing pyrolysis temperature, indicating gradual nitrogen loss during thermal treatment. However, the electrochemical performance does not follow the same trend as the total nitrogen content. For example, NCF-600 possesses the highest overall nitrogen content, yet its capacitance is lower than that of NCF-900. This observation suggests that the chemical state of nitrogen, rather than the total nitrogen content alone, plays an important role in determining electrochemical activity.

We also agree that the specific surface area contributes significantly to charge storage. Indeed, NCF-900 exhibits a higher BET surface area and a more developed mesoporous structure than the other samples, which facilitate electrolyte accessibility and ion transport. Nevertheless, the electrochemical behavior cannot be fully explained by the BET results alone. For instance, the decrease in capacitance from NCF-900 to NCF-1000 is accompanied not only by changes in pore characteristics but also by a substantial reduction in the content of electrochemically active pyridinic-N species.

Therefore, we interpret the superior electrochemical performance of NCF-900 as arising from the synergistic combination of favorable nitrogen configurations, particularly pyridinic-N, together with a defect-rich framework and well-developed mesoporosity. The manuscript has been revised to clarify that pyridinic-N represents a major electrochemically active nitrogen configuration, while the overall device performance is governed by the combined effects of nitrogen chemistry and pore structure.

Table 2. Relative fractions (%) of nitrogen configurations (pyridinic-N, pyrrolic-N, graphitic-N, and oxidized-N) obtained from high-resolution N 1s XPS spectra of NCF-T samples.

Sample	Pyridinic-N (%)	Pyrrolic-N (%)	Graphitic-N (%)	Oxidized-N (%)
Cotton Fabric	-	-	-	-
NCF-600	42.62	35.57	17.27	4.54
NCF-700	40.58	32.57	21.81	5.04
NCF-800	29.89	33.54	22.84	13.73
NCF-900	36.15	32.9	23.57	7.38
NCF-1000	20.34	36.63	29.45	13.58

Table 3. Calculated atomic contents (at.%) of different nitrogen configurations obtained by combining the total nitrogen content from XPS survey spectra with the relative fractions derived from high-resolution N 1s spectra.

Sample	Total N content (at.%)	Calculated Pyridinic-N content (at.%)	Calculated Pyrrolic-N content (at.%)	Calculated Graphitic-N content (at.%)	Calculated Oxidized-N content (at.%)
NCF-600	15.34	6.54	5.46	2.65	0.70
NCF-700	9.05	3.67	2.95	1.97	0.46
NCF-800	8.32	2.49	2.79	1.90	1.14
NCF-900	8.37	3.03	2.75	1.97	0.62
NCF-1000	2.01	0.41	0.74	0.59	0.27

On page 4, Section Introduction of the revised manuscript, we have revised the sentences as follows:

"The sample prepared at 900 °C (NCF-900) exhibited the most favorable combination of a high density of electrochemically accessible pyridinic-N sites, defect-rich structure, and mesoporosity, resulting in superior capacitive performance and durability."

On page 8, Section 3.2 of the revised manuscript, we have revised and added the sentences as follows:

"The relative percentages of nitrogen configurations in the NCF-T series are summarized in Table 2, while the corresponding total nitrogen contents and calculated atomic contents of each nitrogen configuration are presented in Table 3. As the pyrolysis temperature increases from 600 to 1000 °C, the total nitrogen […] Notably, the calculated pyridinic-N content alone does not fully explain the electrochemical performance, indicating that the accessibility and chemical environment of nitrogen species are also important factors governing charge storage. At the same time, a gradual rise in graphitic nitrogen could contribute to enhanced electronic conductivity [21]."

On page 13, Section 3.2 of the revised manuscript, we have revised and added the sentences as follows:

"NCF-900 displayed a smaller semicircle in the high-frequency region, implying lower charge-transfer resistance, and a steeper slope in the low-frequency region, characteristic of efficient ion diffusion [34]. The superior performance of NCF-900 is attributed to the synergistic combination of favorable nitrogen configurations, defect-rich carbon domains, and well-developed mesoporosity, which collectively enhance ion accessibility and charge-storage capability. After 10,000 charge–discharge…"

On page 14 – 15, Section 3.2 of the revised manuscript, we have revised and added the sentences as follows:

"Although NCF-600 exhibits a higher pyridinic-N fraction by XPS, TGA and XPS analyses suggest that a significant portion of these species originate from incompletely decomposed g-C_3N_4 rather than from electrochemically active lattice-incorporated configurations. Consequently, the apparent high pyridinic-N fraction does not directly translate into enhanced charge-storage performance. This observation further indicates that the electrochemical behavior does not simply follow the trend of total nitrogen content or the relative abundance of a specific nitrogen species alone. Instead, the electrochemical activity is governed by the combined effects of nitrogen configuration, nitrogen accessibility, and the structural characteristics of the carbon framework. For NCF-1000…"

On page 15, Section 3.2 of the revised manuscript, we have revised and added the sentences as follows:

"In addition, excessive graphitization at 1000 °C suppresses defect-rich edge sites and reduces ion-accessible active regions. The partial loss of mesoporosity further restricts electrolyte access to electrochemically active sites, resulting in lower pseudocapacitive charge-storage capability despite improved graphitic ordering."

The discussion in the Abstract, Introduction, Sections 3.1, 3.2, 3.3, and the Conclusions has been revised to clarify that the electrochemical performance is governed by the synergistic effects of nitrogen configuration, defect structure, and mesoporosity. In particular, pyridinic-N is discussed as a major electrochemically active nitrogen configuration, while the overall electrochemical performance is interpreted as the combined result of nitrogen chemistry and structural characteristics.